CHAPTER

22 Day case surgery

Learning objectives

To understand:

- The key components of the day surgery pathway
- Which surgical procedures can be done as day surgery
- Patient selection and preparation for day surgery
- Basic principles of anaesthesia and surgery for day surgery
- How to achieve successful discharge after day surgery

DAY SURGERY

In the UK the definition of day surgery is the admission of selected patients to hospital for a planned surgical procedure, returning home on the same day. 'True day surgery' patients are day case patients who require full operating theatre facilities and/or a general anaesthetic, and any day cases not included as outpatients or undergoing endoscopy. Surgery that requires a 23-hour stay, including an overnight stay, is *not* classed as day surgery.

Day surgery offers benefits for patients and hospitals. Patients often prefer to recover in the comfort of their own home, and day surgery may cause less disruption to their domestic situation. It also reduces their risk of a hospital-acquired infection. For the hospital, it can provide greater patient satisfaction and increase the number of inpatient beds available for patients who need to be cared for in hospital.

Successful delivery of day surgery requires the day surgery service to be considered a priority by the hospital, with key enablers in all areas of the pathway providing effective implementation, refinement and progression. There must be a high-quality pathway (***Figure 22.1***) staffed by experienced/expert members of the multidisciplinary team with the equipment and resources they need. This will ensure that there is a well-prepared patient who is in receipt of high-quality day case anaesthesia and surgery and who subsequently has a safe and successful day case discharge.

SELECTION CRITERIA

Surgical

Surgical techniques have progressed significantly and now cause less physiological disruption and stress to patients; therefore, they have a lower postoperative complication profile and a faster recovery rate. The British Association of Day Surgery's (BADS) Directory of Procedures (DOP) lists over 200 procedures that are now considered to be suitable as a day case (***Table 22.1***).

Traditionally day surgery was limited to cases that lasted less than 1 hour but surgical procedures lasting 3–4 hours are now being routinely performed as successful day cases.

Day surgery surgical criteria include the following:

- There must be a low risk of significant immediate postoperative complications, e.g. catastrophic bleeding or airway compromise.
- The patient should be able to eat and drink or take oral nutrition postoperatively.
- Postoperative pain needs to be managed by oral painkillers, which may be in conjunction with local anaesthetic infiltration or peripheral nerve block.
- The patient should be able to mobilise postoperatively with or without aid.

If these criteria are met then the surgeon booking the procedure should add the patient to a day surgery pathway.

TABLE 22.1 Examples from British Association of Day Surgery *Directory of Procedures*, 6th edn (2019).

Specialty	Procedure	Recommended day case rate (%)
Breast	Simple mastectomy	75
Ear–nose–throat	Tonsillectomy	90
General surgery	Laparoscopic cholecystectomy	75
Gynaecology	Vaginal hysterectomy	60
Orthopaedics	Arthroscopy of knee or shoulder	99
Urology	Ureteroscopic extraction of calculus from the ureter	70
Vascular surgery	Transluminal operations on the iliac and femoral arteries	85

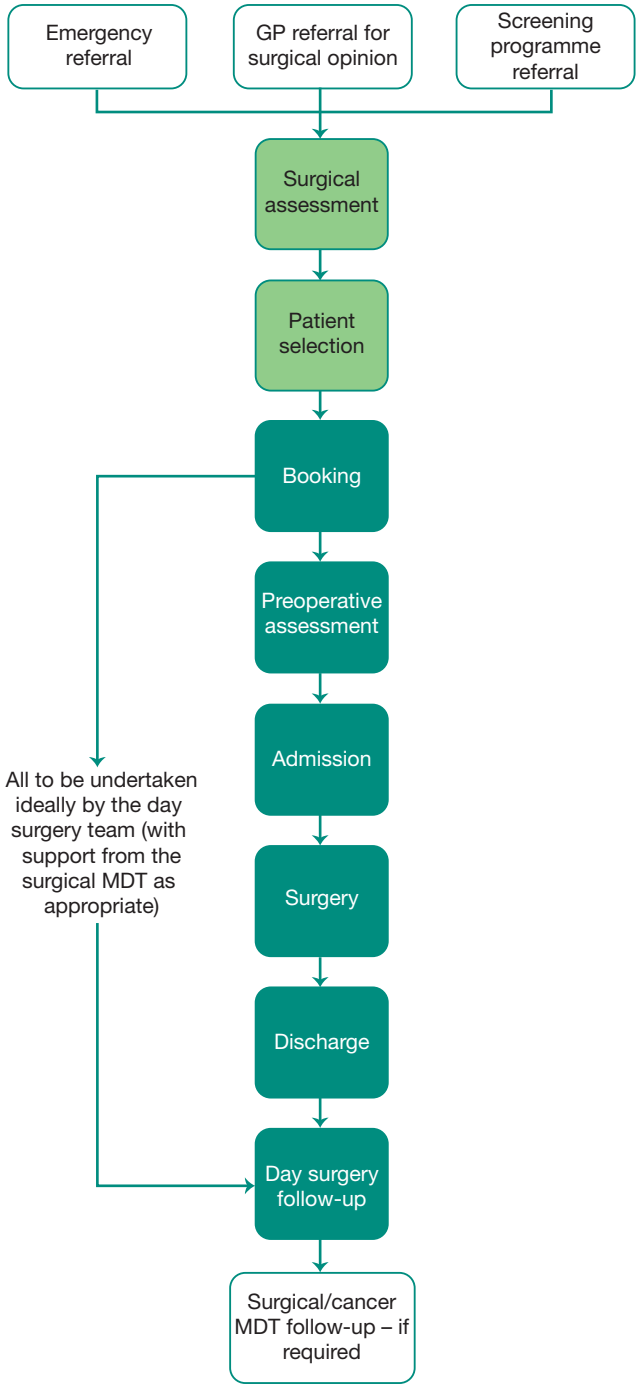


Figure 22.1 Day surgery pathway. GP, general practitioner; MDT, multidisciplinary team.

Medical

With the developments of anaesthesia and surgery, there should be very few restrictions to patients having day surgery (*Table 22.2*). Every effort should be made to optimise a patient's health so that they can be treated as a day case.

There should be no arbitrary cut-offs according to age, weight or criteria specified by the American Society of Anesthesiologists. A patient's suitability for day surgery should be judged on their comorbidities and functional status. Older patients and patients with higher body mass index (BMI) benefit from awake surgery or short-acting anaesthetic agents with a good recovery profile.

TABLE 22.2 Medical exclusions to day surgery.

- Unstable ASA 3
- ASA 4 or 5
- Any poorly controlled abnormality/comorbidity

ASA, American Society of Anesthesiologists.

Diabetes

Patients with diabetes are often better at managing their own diabetes than healthcare professionals. UK national guidance recommends that patients with well-controlled diabetes (haemoglobin A1c [HbA1c] <69 mmol/mol) can be safely managed as a day case. Patients with poorly controlled diabetes have an increased risk of cardiovascular complications and poor wound healing. They should have their surgery delayed until their diabetes is well controlled. If surgery cannot wait or it is thought the underlying disorder (e.g. tooth infection) is causing the diabetes control to be disrupted then diabetic control should be optimised as much as possible prior to surgery.

Epilepsy

Patients with well-controlled epilepsy should not be excluded from day surgery. It is essential that normal medications are not missed. Poorly controlled epilepsy should be optimised prior to any elective surgery.

Obesity

Traditionally there has been caution treating patients who have a higher BMI as a day case. Guidance from the Association of Anaesthetists of Great Britain and Ireland/BADS in 2019 states that 'even morbidly obese patients can be safely managed in expert hands, with appropriate resources'.

Preoperative assessment of patients should routinely include STOP-BANG (Snoring, Tiredness, Observed apnoeas, Pressure [hypertension], Body mass index, Age, Neck circumference, Gender) to identify undiagnosed OSA (obstructive sleep apnoea). The Society for Obesity and Bariatric Anaesthesia (SOBA) Guideline for Anaesthesia of the obese patient identifies a number of risk factors that may make day surgery unsuitable, e.g. poor functional capacity, oxygen saturation <94% on air, STOP-BANG ≥5 (*Figure 22.2*; see also tools.farmacologiaclinica.info, riskcalculator.facs.org/RiskCalculator and www.stopbang.ca). Obese patients considered suitable for day surgery should receive a short-acting anaesthetic, avoiding long-acting opiates, with allowance for the additional time that may be required anaesthetically, surgically and for recovery.

Social

Social criteria for day surgery include:

- Adequate housing conditions such as heating, an inside toilet and access to a phone.

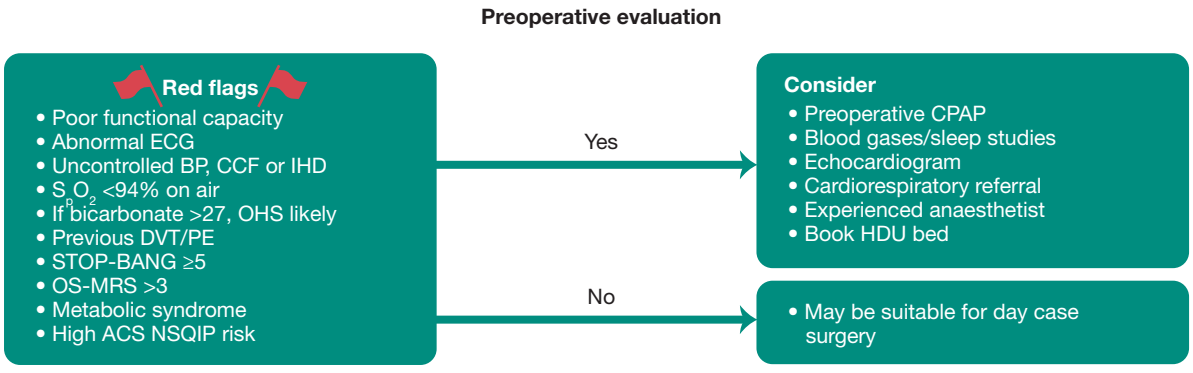


Figure 22.2 Society of Bariatric Anaesthesia (SOBA) red flags. BP, blood pressure; CCF, congestive cardiac failure; CPAP, continuous positive airway pressure; DVT, deep vein thrombosis; ECG, electrocardiogram; HDU, high-dependency unit; IHD, ischaemic heart disease; ACS NSQIP, American College of Surgeons National Surgical Quality Improvement Program; OHS, obesity hypoventilation syndrome; OS-MRS, obesity surgery mortality risk score; PE, pulmonary embolism; S_pO_2, oxygen saturation; STOP-BANG, Snoring ,Tiredness, Observed apnoeas, Pressure (hypertensive), Body mass index, Age, Neck circumference, Gender.

- The patient should live within a 1-hour drive of a hospital.
- A responsible adult should be able to stay with the patient for 24 hours after a regional anaesthetic/general anaesthetic.

The first two points are generally achievable as the patient needs to be 1 hour from 'a hospital' that can treat them rather than the hospital where surgery was performed.

With respect to the responsible adult, there have been two solutions introduced for this by centres in the UK:

1. The Torbay and South Devon NHS Foundation Trust model provides carers into the patient's home.
2. Norfolk and Norwich University Hospital model allows some patients home without carers after certain procedures (*Figure 22.3*).

Both pathways have been in place for a number of years with excellent patient satisfaction and no adverse outcomes.

DELIVERY OF DAY SURGERY

Facilities

National guidance from the Royal College of Anaesthetists, Royal College of Surgeons and BADS recommends that, ideally, day surgery should be performed in a dedicated unit with its own admission area, operating theatres and discharge ward. As a minimum a dedicated day surgery ward is required. This offers a number of benefits, as listed in *Table 22.3*.

It is important to remember that to deliver high-quality successful day surgery the appropriate equipment, drugs and expertise are essential.

TABLE 22.3 Benefits of dedicated day surgery facilities.

- All members of the multidisciplinary team are focused on day surgery
- Nurses with expertise in day surgery
- Nurses not distracted by inpatients
- Activity can continue even during a time of pressures on inpatient beds
- Fewer cancellations because activity can continue even when there are pressures on inpatient beds
- Can be made a COVID secure area – protected from COVID-positive areas of hospital
- Higher chance of successful day case discharge
- Separation from inpatient activity and so patients are more likely to be motivated to get up and go home if they see this as the 'norm'
- Higher patient satisfaction
- Higher quality outcomes

COVID, coronavirus disease.

Preoperative assessment

A key component to successful day surgery is a well-informed, well-prepared patient. It is essential that the day surgery message starts at the time of referral by the primary care doctor and continues throughout the pathway by all staff who the patient interacts with.

Preoperative assessment should follow the same principles as for any other patient and should be nurse led (see *Chapter 21*). The anaesthetist should review the patient's notes where appropriate and the suitability of the patient for day case surgery should be discussed with the day surgery lead to optimise day case rates.

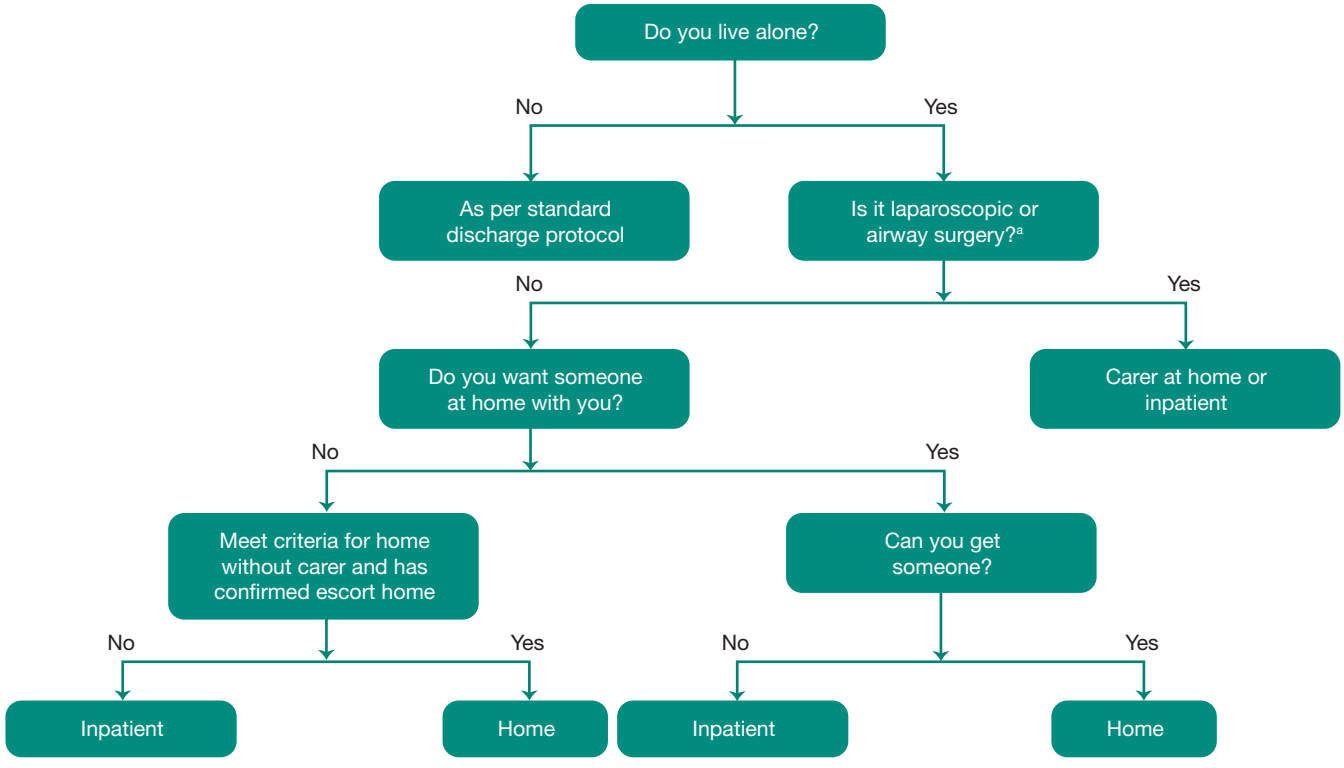
Key preassessment considerations specific to day surgery include:

- Can surgery be delayed until the medical condition is optimised and then plan as a day case?
- Can social factors be addressed for the patient to become a suitable day case?

Admission and list planning

Day surgery patients should follow the same starvation guidance as any other elective patient. All patients, but especially day surgery patients, should be encouraged to walk to theatre.

Consider the list order to optimise successful day surgery and therefore put operations with longer recovery times or patients who take longer to recover early on the lists



All patients must have a responsible adult escort for the journey home

[a]Airway surgery includes nasal and neck procedures and other surgery that may cause bleeding or swelling around the airway

Please discuss with an anaesthetist if the patient has multiple comorbidities or you have any concerns

Figure 22.3 The Norfolk and Norwich Day Surgery Home Alone flowchart. (Reproduced with the permission of The Norfolk and Norwich Day Surgery Team.)

(*Table 22.4*). This needs to be balanced with patients who would benefit from being first on the list, such as patients with insulin-dependent diabetes and patients with learning difficulties who would struggle to wait for long periods of time.

Anaesthesia and surgery

It is not expected that there should be any difference in surgical technique. Surgeons should perform their usual operation, which should be appropriate for rapid recovery and should be performed well. Drains should generally be avoided or, if used, clear plans of when they should be removed and by whom made clear. Any specific postoperative care or discharge information should be documented in theatre to avoid delay to discharge.

Appropriate day surgery anaesthesia requires meticulous attention to ensuring good pain relief and avoidance of postoperative nausea and vomiting. This should include premedication and a multimodal approach. Short-acting general anaesthesia agents, day case spinals or regional anaesthesia techniques should be used to enable rapid recovery. Use of long-acting opioids such as intravenous morphine is discouraged because they can delay recovery owing to increased sleepiness or nausea.

TABLE 22.4 List planning.

Operation with potentially longer recovery times	Types of patients who might need longer recovery time
Tonsillectomy	Very elderly
Knee replacement	High BMI
Hip replacement	
Complex laparoscopic cholecystectomy	

BMI, body mass index.

Discharge

The expectation by the patient and healthcare team should be that the patient will be going home the same day. Therefore, unless there has been an unexpected anaesthetic or surgical issue the patient should routinely have a nurse-led discharge. Patients should meet any pre-agreed general criteria (*Table 22.5*) as well as any surgery-specific criteria prior to discharge. In general, there should be no time restriction except for certain procedures, e.g. patients should remain in hospital until 6 hours after tonsillectomy.

TABLE 22.5 Discharge criteria.

- Vital signs stable for at least 1 hour
- Correct orientation as to time, place and person if appropriate
- Adequate pain control with supply of oral analgesia
- Understands how to use oral analgesia supplied
- Ability to dress and walk where appropriate
- Minimal nausea, vomiting or dizziness
- Has taken oral fluids
- Minimal bleeding or wound drainage
- Has passed urine (if appropriate)
- Has a responsible adult to take them home
- Written and verbal instructions given about postoperative care
- Knows when to come back for follow-up (if appropriate)
- Emergency contact number supplied

The patient should receive written and verbal postoperative instructions and a phone number to contact should they have a problem out of hours. This must be a phone with a suitable person to advise and not an answerphone.

Take-home medications should provide adequate pain relief and may include an antiemetic. These should be prescribed when the patient is in theatre and pre-packs of common analgesics should be used to improve the efficiency of prescribing and reduce delays to discharge.

All day surgery patients should be telephoned the day after surgery to provide support and to check that they have no problems. This call can also be used to collect valuable audit data, which can be used to refine the day surgery pathway.

EMERGENCY DAY SURGERY

Many emergency surgical procedures are minor and non-life-threatening. Patients may be considered low priority for surgical intervention and can therefore end up waiting hours or days for a slot on the emergency theatre list, resulting in prolonged starvation times and inpatient stay.

With appropriate planning and preparation these patients could have surgery performed as a day case. This has become reasonably commonplace in orthopaedics for many upper limb traumas and in gynaecology for the evacuation of retained products of conception (ERPC). It has also increasingly been recognised for many other surgical procedures, as listed in the BADS DOP (***Table 22.6***).

For certain procedures that can wait more than 24 hours patients can follow an 'elective pathway'. They can be swabbed and isolate as per current coronavirus 2019 (COVID-19) requirements and then attend via an 'elective green pathway'. Alternatively, they can be discharged home and then return to an acute surgical admission area to be added to a suitable list or be first on the emergency list (priority slot) and discharged the same day.

Contraindications to being discharged must be identified, e.g. systemic sepsis, unstable diabetes, major comorbidities, if parenteral pain relief is needed or if patients are deemed unsafe to mobilise.

TABLE 22.6 Common procedures suitable for an emergency day surgery pathway.

Procedure	Suggested BADS DOP day case rate (%)
Evacuation of retained products of conception	95
Incision and drainage of a perianal abscess	95
Appendicectomy	15
Reduction of a fracture of the zygomatic complex of bones	60
Repair of hand or wrist tendon	95
Primary reduction and open fixation of the ankle	25
Primary reduction and open fixation of the wrist	60

BADS DOP, British Association of Day Surgery *Directory of Procedures*, 6th edn (2019).

FURTHER READING

Bailey CR, Ahuja M, Bartholomew K *et al.* Guidelines for day-case surgery 2019: guidelines from the Association of Anaesthetists and the British Association of Day Surgery. *Anaesthesia* 2019; **74**(6): 778–92.

British Association of Day Surgery. *BADS directory of procedures*, 6th edn, 2019. Available from https://publications.bads.co.uk.

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Russon K *et al. Chapter 6 Guidelines for the provision of anaesthesia services for day surgery*, 2020. Available from https://rcoa.ac.uk/gpas/chapter-6.

Stocker M *et al. National day surgery delivery pack*, 2020. Available from https://www.gettingitrightfirsttime.co.uk/bpl/day-surgery/.

British Association of Day Surgery booklets (www.bads.co.uk):
Day case breast surgery (2020)
Day case gynaecology (2020)
Day case hip & knee replacement, 2nd edn (2020)
Day case laparoscopic cholecystectomy, 3rd edn (2018)
Managing diabetes in patients having day and short stay surgery, 4th edn (2016)
Nurse led discharge, 2nd edn (2016)
Spinal anaesthesia for day surgery patients: a practical guide, 4th edn (2019)
Surgical same-day emergency care, 2nd edn (2020)